

Engineering Resolution Report: LOW-01 through LOW-08

Legal Analyzer Platform • Date: October 10, 2026 • Status: All 8 Issues Resolved & Verified

1. Executive Summary Matrix

Issue ID	Architecture Label	Domain	Target File / Component	Status
LOW-01	CODE-STYLE-001-ELIMINATE-PRINTS	Observability & Style	Document routers, storage & OCR services	RESOLVED
LOW-02	CODE-QUALITY-001-REMOVE-DEAD-CHECKS	Code Quality	Document sub-routers (crud, analysis, review, chat)	RESOLVED
LOW-03	SEC-HDR-001-PERMISSIONS-POLICY	Security Hardening	backend/app/core/security_header.s.py	RESOLVED
LOW-04	DEVOPS-COMPOSE-001-CLEANUP	DevOps & Containers	docker-compose.yml, backend/Dockerfile	RESOLVED
LOW-05	DEVOPS-DEP-001-PIN-VERSIONS	Dependencies	backend/requirements.txt	RESOLVED
LOW-06	LEGAL-LIC-001-ADD-LICENSE	Legal & Governance	Root LICENSE file (MIT)	RESOLVED
LOW-07	DATA-DB-007-CHUNK-INDEX-CONSTRAINT	Database Integrity	DocumentChunk & Alembic migration	RESOLVED
LOW-08	DOCS-META-001-URL-CLEANUP	Documentation Meta	README.md repository meta & links	RESOLVED

2. Technical Problem, Resolution & Verification Details

LOW-01: Replace Lingering Print Statements with Structured Logger

Label: CODE-STYLE-001-ELIMINATE-PRINTS • **Target:** backend/app/api/v1/routers/documents/*, storage_service.py, ocr_service.py

Problem: Raw print() statements in router helpers and core services outputted unformatted text to standard out, bypassing JSON formatters, correlation ID (X-Request-ID) tracking, and Sentry breadcrumbs.

Solution: Initialized module loggers (logger = logging.getLogger(__name__)). Replaced all print() statements with logger.info() and logger.warning(). Enforced document confidentiality by logging strictly document_id references.

```
logger = logging.getLogger(__name__)
logger.info('Storage service initialized with provider: %s', provider)
logger.warning('OCR processing warning for document %s: %s', document_id, err)
```

Verification: Unit tests in tests/test_logging.py passed; codebase scan confirmed 0 unmanaged print() statements.

LOW-02: Eliminate Dead Code Unreachable Null Checks

Label: CODE-QUALITY-001-REMOVE-DEAD-CHECKS • **Target:** backend/app/api/v1/routers/documents/crud.py, analysis.py, review.py, chat.py

Problem: get_accessible_document() raises an HTTPException(404) if a document is not found. Immediate subsequent checks (if not doc: raise HTTPException(...)) across endpoints were unreachable dead code.

Solution: Audited and eliminated 6 blocks of unreachable dead null checks across documents sub-routers, streamlining control flow and eliminating branch coverage gaps.

```
# Cleaned endpoint invocation:
doc = get_accessible_document(db, document_id, current_user)
# Dead 'if not doc:' checks removed across all sub-routers
```

Verification: ruff check app/api/v1/routers/documents/ passed with 0 warnings; endpoints properly return 404.

LOW-03: Inject Permissions-Policy Header in Security Middleware

Label: SEC-HDR-001-PERMISSIONS-POLICY • **Target:** backend/app/core/security_headers.py, backend/tests/test_security_headers_and_rate_limit.py

Problem: Security headers middleware lacked a Permissions-Policy HTTP header, leaving browser hardware APIs (camera, microphone, geolocation, payment) unrestricted.

Solution: Injected Permissions-Policy header in SecurityHeadersMiddleware.dispatch() with strict zero-permission rules: camera=(), microphone=(), geolocation=(), payment=().

```
response.headers['Permissions-Policy'] = 'camera=(), microphone=(), geolocation=(), payment=()'
```

Verification: pytest tests/test_security_headers_and_rate_limit.py verified that all HTTP responses carry the Permissions-Policy header.

LOW-04: Modernize docker-compose.yml and Fix Volume Pre-Creation Dependency

Label: DEVOPS-COMPOSE-001-CLEANUP • **Target:** docker-compose.yml, backend/Dockerfile

Problem: docker-compose.yml specified deprecated version: '3.8' and external: true on n8n_data, failing docker compose up if volume was absent. Dockerfile lacked a container HEALTHCHECK.

Solution: Removed version: '3.8', converted n8n_data to a self-managed volume (n8n_data:), and added a HEALTHCHECK instruction targeting http://localhost:8000/health in backend/Dockerfile.

```
HEALTHCHECK --interval=30s --timeout=5s --start-period=10s --retries=3 \
  CMD python -c "import urllib.request; urllib.request.urlopen('http://localhost:8000/health')" || exit 1
```

Verification: docker-compose file syntax validated with 0 schema warnings.

LOW-05: Pin Backend Dependencies with Upper Version Bounds

Label: DEVOPS-DEP-001-PIN-VERSIONS • **Target:** backend/requirements.txt

Problem: Dependencies in requirements.txt used unbounded >= minimum versions, leaving builds and CI workflows vulnerable to unexpected breaking changes from upstream major releases.

Solution: Pinned all package specifiers with compatible upper version bounds (e.g., fastapi>=0.110.0,<1.0.0, pydantic>=2.6.0,<3.0.0, sqlalchemy>=2.0.0,<3.0.0).

```
fastapi>=0.110.0,<1.0.0
pydantic>=2.6.0,<3.0.0
sqlalchemy>=2.0.0,<3.0.0
alembic>=1.13.0,<2.0.0
reportlab>=4.1.0,<5.0.0
```

Verification: pip check verified consistent dependency graph with no version conflicts.

LOW-06: Add Root LICENSE File

Label: LEGAL-LIC-001-ADD-LICENSE • **Target:** LICENSE (Root repository)

Problem: README.md stated the project was released under the MIT License, but there was no LICENSE file in the repository root, creating legal ambiguity for open-source consumers.

Solution: Created official MIT License file at repository root with standard legal terms and 2026 copyright attribution to Kushagra.

```
MIT License
Copyright (c) 2026 Kushagra
Permission is hereby granted, free of charge...
```

Verification: Verified file presence at root; complies with standard SPDX MIT format.

LOW-07: Enforce Unique Constraint on Document Chunk Indexes

Label: DATA-DB-007-CHUNK-INDEX-CONSTRAINT • **Target:** backend/app/models/database_models.py, alembic migration, backend/tests/test_embedding_and_chunks.py

Problem: Mid-process chunking failures during retries could persist duplicate chunk rows with identical index numbers for the same document, degrading embedding quality and RAG retrieval.

Solution: Added UniqueConstraint('document_id', 'chunk_index', name='uq_document_chunk_index') to DocumentChunk.__table_args__. Generated Alembic migration 2e1482525d09, and added unit test.

```
from sqlalchemy import UniqueConstraint
__table_args__ = (
    UniqueConstraint('document_id', 'chunk_index', name='uq_document_chunk_index'),
)
```

Verification: pytest tests/test_embedding_and_chunks.py passed 6/6 tests in 23.38s; verified duplicate insert raises IntegrityError.

LOW-08: Update Placeholder Badge URLs and Documentation Meta

Label: DOCS-META-001-URL-CLEANUP • **Target:** README.md

Problem: README.md contained placeholder URLs (your-username/legal-analyzer, your-email@example.com), outdated placeholder assumption notes, and dead image references to uncommitted files.

Solution: Updated all badges and clone commands to Kushagra901/Legal-analyzer. Removed dead image references, replacing them with structured UI descriptions. Updated maintainer contact info and fixed appendix links.

```
![Build Status](https://img.shields.io/github/actions/workflow/status/Kushagra901/Legal-analyzer/ci.yml?branch=main)
**Primary maintainer:** Kushagra (@Kushagra901)
Architecture doc: legal-analyzer-architecture.md
```

Verification: Grep scan confirmed 0 placeholder URLs (your-username, your-email, your-name, example.com) remain in README.md.

3. Quality & Test Execution Metrics

Verification Scope	Execution Command	Results
Chunk Unique Constraint & Embeddings	pytest tests/test_embedding_and_chunks.py	6 passed (100% success)
Security Headers & Permissions-Policy	pytest tests/test_security_headers_and_rate_limit.py	5 passed (100% success)
Logging & Zero Raw Prints Check	pytest tests/test_logging.py	8 passed (100% success)
Python Code Linter	ruff check app tests alembic	0 errors (Clean)